



An examination of stability in cohesive zone modeling

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ABSTRACT

Stable crack propagation hinges on the driving force and the resistance. In the context of a cohesive approach to fracture, properly resolving the cohesive zone ensures that the resistance is in equilibrium with the driving force. Additional restrictions on the mesh size can stem from crack stability. For high strength, low toughness (brittle) materials, the requirements for stability can exceed those for cohesive zone resolution. Examples in 1-D and 2-D reveal that decrements in the mesh size transition the system from indefinite to positive definite. Moreover, small decreases in the mesh size beyond the transition provide substantial reductions in the condition number. We contrast a physical instability resulting from material properties, geometry, or boundary conditions from a numerical instability resulting from the mesh size and propose that the selected discretization should ensure that the cohesive zone is both resolved and stabilized.

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1. Introduction

Cohesive surface methods [1,6] provide a phenomenological representation of the fracture process and yield regularized propagation. Early works focused on interfaces [11–13,19], cohesive networks [21], and adaptive insertion [4]. Please refer to [10] for a review of initial studies. In this work, we focus on the quasi-static case and do not consider the physics underlying dynamic instabilities. Our intent is to examine numerically-induced instabilities for idealized cases of extension in 1-D and three-point bending in 2-D. Instability is defined in terms of the driving force J [7,15] and the resistance R . During the propagation of a crack of length a , the driving force is in equilibrium with the resistance, $J = R$. The process will be stable provided $\partial J/\partial a < \partial R/\partial a$ [17].

[5] and [2,3] examined snap-back instabilities associated with the three-point bend geometry. Depending on the material properties, geometry, and loading, behavior was described as “brittle” (snap-back) or “ductile.” In contrast to large cohesive zones that promote “ductile” behavior, the cohesive zones in this work are small compared to all dimensions of the body. Provided the cohesive zone is well resolved and the bulk is linear elastic, the only cohesive property of interest is the fracture energy [15] and one can align propagation with linear elastic fracture mechanics (LEFM). This work will illustrate that even if the cohesive zone is well resolved, further decrements in the mesh size may be necessary to ensure stability. Just as propagation $J = R$ and stability

$\partial J/\partial a < \partial R/\partial a$ are independent quantities, the required discretization for resolution and stability are also independent quantities.

2. Implementation

2.1. Constitutive models

In the current study, we partition the material response into bulk and surface constitutive models. To simplify arguments and facilitate comparison to analytical solutions, we restrict this study to infinitesimal deformations and assume the bulk to be isotropic and linear elastic having properties expressed through Young's modulus E and Poisson's ratio ν or the Lamé elastic constants λ and μ . Cohesive surface relations are taken from [21]. For this work, we postulate that the normal fracture energy ϕ_n and the shear fracture energy ϕ_t are equivalent $\phi = \phi_n = \phi_t$. The resulting components of the traction are

$$T_n(\Delta) = \phi \frac{\Delta_n}{\delta_n^2} \exp\left(-\frac{\Delta_n}{\delta_n}\right) \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right) \quad (1)$$

and

$$T_t(\Delta) = 2 \phi \frac{\Delta_t}{\delta_t^2} \left(1 + \frac{\Delta_n}{\delta_n}\right) \exp\left(-\frac{\Delta_n}{\delta_n}\right) \exp\left(-\frac{\Delta_t^2}{\delta_t^2}\right). \quad (2)$$

where δ_n and δ_t are the characteristic length scales governing normal and tangential separation, respectively. The normal and tangential openings are defined as $\Delta_n = \Delta \cdot \mathbf{n}$ and $\Delta_t = \Delta \cdot \mathbf{t}$ with respect to the local normal $\mathbf{n}$ and tangent $\mathbf{t}$ to the cohesive surface element. Because the form of the separation law is specified, one

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can find the normal strength σ_{max} and shear strength τ_{max} through the fracture energy ϕ and characteristic lengths δ_n, δ_t .

2.2. Bulk and cohesive element tangent stiffness

To obtain a better understanding of the effect of element size on stability, we investigate the quasi-static, normal separation of two bulk elements of size h joined by a cohesive surface element. A schematic of the problem is illustrated in Fig. 1. Using the formulation outlined by Klein et al. [10], we can derive the tangent stiffness matrix for the cohesive surface element and assemble the global stiffness matrix. We begin in 1-D with linear basis functions and integrate to find the bulk K^{el}

$$K^{el} = \begin{bmatrix} \frac{E}{h} & -\frac{E}{h} \\ -\frac{E}{h} & \frac{E}{h} \end{bmatrix} \quad (3)$$

and cohesive surface K_{cs}^{el} element stiffness matrices

$$K_{cs}^{el} = \begin{bmatrix} \frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} & -\frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} \\ -\frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} & \frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} \end{bmatrix} \quad (4)$$

for unit area. After assembling element stiffness matrices and applying boundary conditions, we can express the global stiffness

$$K = \begin{bmatrix} \frac{E}{h} + \frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} & -\frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} \\ -\frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} & \frac{E}{h} + \frac{\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2} \end{bmatrix} \quad (5)$$

as a function of material properties $E, \sigma_{max}, \delta_n$, geometry h , and normal opening Δ_n . We can also construct both the bulk and cohesive surface element stiffness contributions in 2-D. For brevity, we only express a diagonal term corresponding to a normal ($\mathbf{e}_1$) degree of freedom of node β (Fig. 1).

$$K(2\beta - 1, 2\beta - 1) = \frac{2}{3}(\lambda + 3\mu) + \frac{2h\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n} - \frac{\Delta_n^2}{\delta_n^2})}{3\delta_n^2} \quad (6)$$

Because bulk stresses and the resulting tractions derive from strain measures and cohesive tractions derive from opening displacements, bulk stiffness terms will always differ from cohesive stiffness terms by $1/h$. Eq. (6) confirms the role of mesh size. The implication of this simple exercise is to illustrate that we can scale

the relative contribution of the cohesive surface elements through mesh size. Subsequent sections will illustrate the impact on stability of the system in 1-D and 2-D.

3. Stability in 1-D

In an effort to keep all expressions analytic, we explore stability for the idealized, 1-D system displayed in Eq. (5). We align stability with a positive definite stiffness matrix. Because a defect a and the corresponding driving force J do not exist, we are unable to make the explicit link between crack stability and stability of the 1-D model system. Nevertheless, the idealization permits simple analytical representations, contains the $1/h$ scaling, and provides a path towards understanding stability in flawed bodies. The eigenvalues of the global stiffness matrix K are

$$\lambda_1(\Delta_n) = \frac{E}{h} + \frac{2\sigma_{max}(\delta_n - \Delta_n) \exp(1 - \frac{\Delta_n}{\delta_n})}{\delta_n^2}, \quad \lambda_2 = \frac{E}{h}, \quad (7)$$

where the first eigenvalue λ_1 is a function of the normal opening Δ_n . In an effort to determine the impact of mesh size and material properties on the stability of the system, we minimize λ_1 with respect to Δ_n and require that $\lambda_1 > 0$, the global stiffness matrix is positive definite. We find that the minimum for the chosen potential occurs at $\Delta_n = 2\delta_n$. The minimum eigenvalue corresponds to the minimum slope in the traction–displacement law. Employing the identity $\phi = \sigma_{max}\delta_n \exp(1)$, and evaluating λ_1 at $\Delta_n = 2\delta_n$ we find

$$\lambda_{1,min} = \frac{E}{h} - \frac{2\sigma_{max}^2}{\phi} \quad (8)$$

which places the following restriction on the mesh size

$$h < \frac{E\phi}{2\sigma_{max}^2} \quad (9)$$

Although qualitative, this 1-D example illustrates a difficulty of employing a cohesive approach to simulate fracture in high strength σ_{max} , low toughness ϕ material systems. The level of discretization required to avoid the snap-back instability can be substantial. The restriction can be clarified through the introduction of a fracture length scale [14]. The length scale, akin to an unscaled plastic zone size, is $r = E\phi/\sigma_{max}^2$. The restriction on the mesh size can now be expressed as $r/h > 2$. Materials of small r , brittle materials, require small mesh sizes to ensure stability.

3.1. Condition number

In addition to placing a limitation on the mesh size, we can also employ the fracture length scale to investigate the maximum condition number $c_{max} = \lambda_2/\lambda_{1,min}$ of the idealized system

$$c_{max} = \frac{1}{1 - \frac{2h}{r}} \quad (10)$$

Fig. 2 illustrates the maximum condition number. As expected, $c_{max} \rightarrow 1$ as $h/r \rightarrow 0$. We also note that small decreases in the mesh size beyond the restriction $r/h > 2$ yield marked improvements in c_{max} . For example, if we refine h by a only a factor of two, we obtain $c_{max} = 2$. The rapid reduction in the condition number for relatively small changes in the mesh size has implications for both the sensitivity of the solution [18] and the rate of convergence of iterative solvers such as those of the Krylov class which include steepest descent and conjugate gradient methods [22].

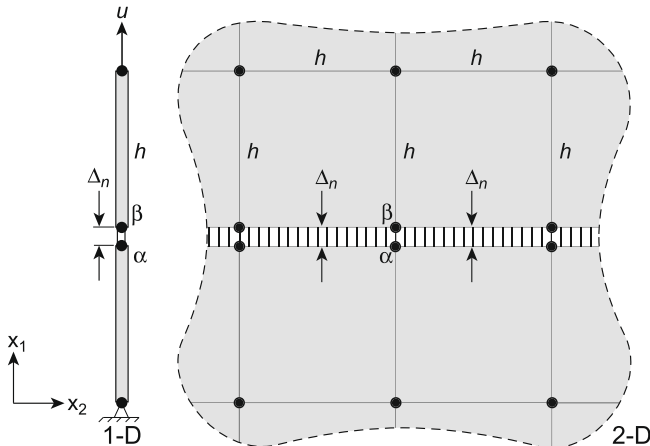


Fig. 1. Schematic of normal separation Δ_n for 1-D and 2-D geometries having a uniform mesh size h . The local normal $\mathbf{n}$ is aligned with $\mathbf{e}_1$. The 1-D case represents a three element structure while the 2-D case is illustrative of the bulk and cohesive surface element contributions to a larger domain.

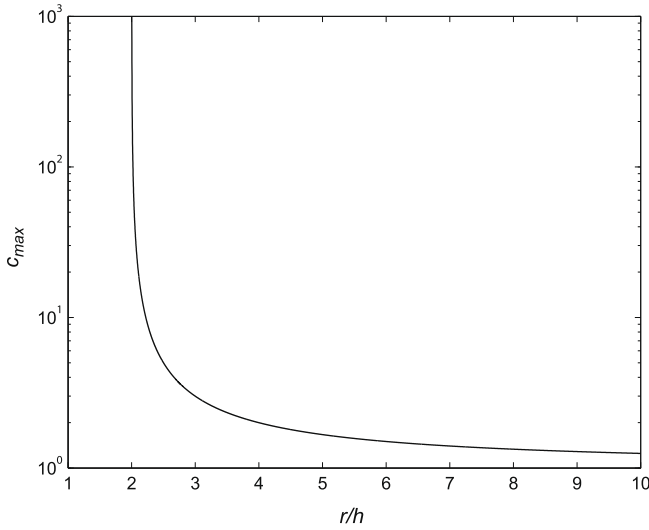


Fig. 2. Variation of maximum condition number c_{max} for mesh size h normalized by the fracture length scale $r = E\phi/\sigma_{max}^2$. Relatively small decreases in the mesh size $h = r/4$ yield rapid reductions in the condition number $c_{max} = 2$.

3.2. Additional cohesive descriptions

Because the minimum eigenvalue is related to the minimum cohesive stiffness, cohesive relations that maximize the unload stiffness through linear decay [4] are the most stable. Given a linear decay from σ_{max} over δ_n , the fracture energy can be expressed as $\phi = \frac{1}{2} \sigma_{max} \delta_n$. For this case, the cohesive surface element stiffness matrix has a simplified form

$$K_{cs,linear}^{el} = \begin{bmatrix} -\frac{\sigma_{max}}{\delta_n} & \frac{\sigma_{max}}{\delta_n} \\ \frac{\sigma_{max}}{\delta_n} & -\frac{\sigma_{max}}{\delta_n} \end{bmatrix} \quad (11)$$

and the minimum eigenvalue $\lambda_{1,linear}$ resulting from the global stiffness matrix is $E/h - 2\sigma/\delta$ which provides a restriction on the mesh size

$$h_{linear} < \frac{E\phi}{\sigma_{max}^2}, \quad (12)$$

where $h_{linear} = r = 2h$.

Although we have defined stability through the minimum eigenvalue, one could also derive the aforementioned restrictions based on the macroscopic load P and deflection u . Enforcing the inequality $dP/du < 0$ prevents snap-back and yields the restrictions given in Eqs. (9) and (12). Because the stress is constant throughout the bar, the bulk elements must be sufficiently stiff to accommodate the specified unloading in the cohesive surface element. If the elements are not sufficiently small, the only path available would be a reduction in both P and u , $dP/du > 0$, the snap-back instability.

We stress that because a defect a is not present in the 1-D structure, restrictions on the mesh size stem from a physical instability. In 2-D, however, we can examine stable geometries $\partial J/\partial a < \partial R/\partial a$ in light of 1-D findings. In addition to resolving the cohesive zone, one may be forced to further refine the mesh to obtain stable propagation. Refinement is needed because crack propagation is often an unstable process and one should ensure that instabilities result from the loading, geometry, and/or material properties and *not* the selected level of discretization. We note that although the 1-D example is helpful in developing analytical expressions for stability, the simplified case is not representative of stability during propagation. A defect and the corresponding driving force do not exist ($J = 0$). In the next section, we delineate unstable and stable

quasi-static propagation for a crack of length a in the standard three-point bend geometry.

4. Stable and unstable propagation in three-point bending

To obtain a better understanding of stability, we investigate the three-point bend geometry. For a given set of material properties, we seek to determine the mesh size that will enable stable propagation. Based on findings in 1-D, we select a model material that has a small fracture length scale, e.g. ceramic material. The elastic constants in Table 1 are taken from a prior study on grain bridging in brittle materials [8].

The resulting fracture length scale for the model ceramic system is $r = 120$ nm and the corresponding mesh size restriction from 1-D is $h < 60$ nm. Prior to investigating numeric stability, we need to ensure that the geometry is stable. Because we are investigating the motion of a crack of length a , we define physical stability through the driving force J and the resistance R . The specimen (geometry) is stable during crack propagation, $J = R$, provided $\partial R/\partial a > \partial J/\partial a$. Because the bulk is elastic and the crack path is aligned with the mode of loading, R is a constant material parameter ϕ and is not a function of a . Consequently, the specimen is stable if $\partial J/\partial a < 0$. Although the distinction between a physically and numerically unstable system might seem arbitrary, our studies of numeric stability need to be grounded in a physically stable configuration. Because the three-point bend geometry is a standard fracture geometry, we can employ analytical solutions to determine $J = J(a)$ and place restrictions on the initial crack length.

4.1. Analytical solution

To calculate the stability of the specimen, we employ analytical solutions to determine $J = J(a)$ and $\partial J/\partial a$. The applied stress intensity [16] is

$$K = \frac{3PS}{BW^{3/2}} \left(\frac{a}{W} \right)^{1/2} f(a/W) \quad (13)$$

with

$$f(a/W) = \frac{1.99 - a/W(1 - a/W)(2.15 - 3.93a/W + 2.7(a/W)^2)}{2(1 + 2a/W)(1 - a/W)^{3/2}}, \quad (14)$$

where P is the applied load, S is the beam span, B is the beam thickness, a is the crack length, and W is beam width. Because the bulk is linear elastic, we can calculate the driving force $J = K^2/E'$ where $E' = E/(1 - \nu)^2$ for plane strain. We can also employ a load–line displacement relation [20]

$$u = \frac{P}{EB} \left(\frac{S}{W} \right)^2 \frac{g(a/W)}{(1 - a/W)^2}, \quad (15)$$

Table 1

Bulk and interface properties employed for the model brittle system.

Ceramic	Property	Value	Units
Bulk	E	303	GPa
	ν	0.21	–
	ϕ	40	J/m ²
	σ_{max}	10	GPa
Surface	τ_{max}	10	GPa
	δ_n	1.5	nm
	δ_t	3.4	nm

where

$$g(a/W) = 1.193 - 1.980a/W + 4.478(a/W)^2 - 4.443(a/W)^3 + 1.739(a/W)^4 \quad (16)$$

to find the driving force under displacement control. If we fix the displacement u , material properties E, ν , and beam geometry S, B, W , we can express $J = J(a)$. Stability is determined by $\partial J / \partial a|_u$. The driving force has a well defined maximum $J_{\max}$ located at $a/W = 0.31$. The normalized driving force $J/J_{\max}$ for displacement $J(a)|_u$ and load $J(a)|_p$ control is illustrated in Fig. 3. In load control, the driving force is normalized by $J(p, a)$ evaluated at $a/W = 0.8$. As expected, the body is unstable under load control. We also note that the body is unstable in displacement control for $a/W \leq 0.31$. Displacement boundary conditions are not sufficient to guarantee stability. For this study, we choose to model a stable configuration in displacement control, $a/W = 0.5$.

4.2. Numerical study

The purpose of the study is to numerically calculate restrictions on the cohesive surface mesh size h for stable mode I propagation. Often, we select an element size to properly resolve the cohesive zone. However, further mesh refinement may be needed to avoid a singular tangent stiffness matrix. We explore this hypothesis with the three-point bend geometry in displacement control. To facilitate mesh refinement, the beam width W is taken to be $25.4 \mu\text{m}$. The required span $S = 4W$ is $101.6 \mu\text{m}$ and the overall length of the beam is $5W$, $127 \mu\text{m}$. The pre-crack is $12.7 \mu\text{m}$, $a/W = 0.5$. We note that if the cohesive zone is confined, $l_{cz} \ll a$, the analytical solution is valid.

Although the 1-D case is idealized, we begin with $r/h = 2$ where $r = 120 \text{ nm}$ and refine to stability. Representative cases in Fig. 4 reflect mesh refinement for $r/h = 2, 3, 6, 12$, and 24 . The refinement is accomplished via paving algorithms. Each case represents local (not uniform) refinement at the crack tip. Although the methodology is ad hoc, it reflects current practice and is appropriate for illustrative purposes. A direct solver is employed and the applied displacement increment is sequentially cut 20 times ($1/1,000,000$) to instability. Bordering the instability, the maximum and minimum eigenvalues are noted over 20 iterations.

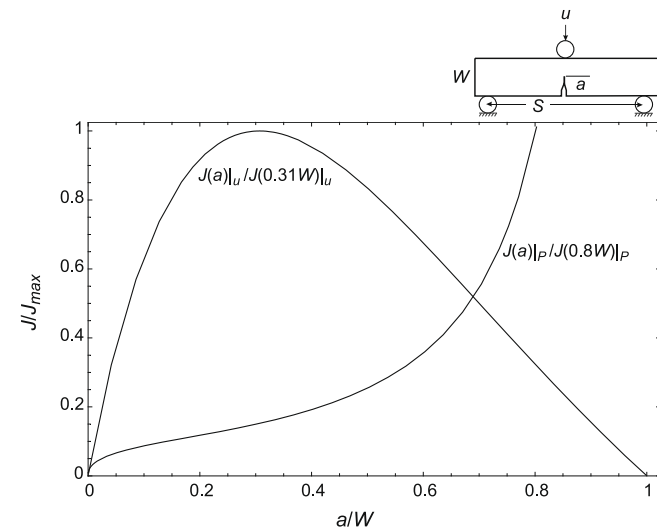


Fig. 3. Calculated driving force for a constant load-line displacement. For a constant resistance, the specimen is stable if $\partial J / \partial a < 0$. In displacement control $J(a)|_u$, the specimen is unstable for $a/W \leq 0.31$. The specimen is always unstable in load control $J(a)|_p$.

Simulations for $r/h = 2, 3$, and 6 do not converge. In contrast, $r/h = 12$ and 24 result in stable propagation. The mesh size is subsequently bisected three times to determine the required level of discretization, $r/h > 7.4$, for stable propagation. Macroscopic curves for stable mesh sizes are illustrated in Fig. 5a. The reaction load P is normalized by the analytical solution, Eq. (13), for $a/W = 0.5$. For all simulations, the crack tip is assumed to coincide with the tail of the global traction distribution $\sim 0.1T_n$. As noted in Fig. 5, all stable mesh sizes are converged. The rapid transition from unstable to stable propagation is noted in Fig. 5b. Changing the mesh size by roughly a nanometer, the system is stable and a solution is obtained.

To determine the degree of cohesive zone resolution, we adopt a conservative estimate of the cohesive zone size. The cohesive zone size l_{cz} spans from $\sigma_{\max}$ to $0.1\sigma_{\max}$ in the unloaded portion of the global traction distribution. We note that for the current potential, $0.1\sigma_{\max}$ corresponds to $0.96\phi_n$. For the finest case $r/h = 64$, $h = 2.5 \text{ nm}$, we obtain $l_{cz} = 52 \text{ nm} = 0.43r$. Clearly, all stable meshes $h \leq 16.2 \text{ nm}$ are well resolved. We note that during subsequent bisection, $h = 17.5 \text{ nm}$ yields an instability. With roughly three elements in the cohesive zone, the level of discretization for $h = 17.5 \text{ nm}$ is ample for cohesive zone resolution [10]. It is, however, not adequate for stability. For this specific case, the requirements for stability exceed the requirements for resolution. As noted in Fig. 5a, all meshes that are stable are “resolved.” The cohesive zone size also sheds light on the validity of the analytical solution. Because the cohesive zone size is small compared to all dimensions of the body $l_{cz}/a < 1/100$, a K -annulus exists and the analytical solution is valid.

Having qualitatively established the effects of discretization, we can also examine minimum eigenvalues and the impact on the condition number. Rather than normalize by the fracture length scale r , we express measures of stability in terms of the simulated length scale, the process zone size l_{cz} . Fig. 6a illustrates the variance in first eigenvalue λ_1 with the number of elements in the process zone l_{cz}/h . The minimum, mean, and maximum of the first eigenvalue are taken from 20 iterations bordering instability $\{\lambda_1^1 \dots \lambda_1^{20}\}$. The minimum λ_1 indicates a distinct transition from indefinite to positive definite for $l_{cz}/h > 3$. Mean eigenvalues illustrate the system is nearly singular during the iterative process for $l_{cz}/h < 3$. Because the discretization at the crack tip is local, comparison to a bulk mesh of the same discretization is needed to separate the effects of bulk and cohesive contributions. Simulations of sharp cracks (without cohesive surface elements) illustrate that for $l_{cz}/h > 3$ the maximum $\{\lambda_1^i\}$ correlates well with λ_1 from bulk simulations.

The condition number $c = |\lambda_{\max}|/|\lambda_{\min}|$ is also calculated for each iteration $\{c^1 \dots c^n\}$ and the minimum, mean, and maximum condition numbers are noted in Fig. 6b. The mean condition number c rapidly decreases by three orders of magnitude for decreasing mesh size. As expected, the minimum condition number during the iterative process for the stabilized cases aligns with the bulk condition number. Fig. 6 clearly illustrates that refining past the condition for stability $l_{cz}/h > 3$ is not beneficial and that increases in the condition number past the stability condition reflect bulk contributions and are relatively insensitive to cohesive contributions.

In this work, we have emphasized the role of the mesh size in promoting stability. We have only examined a single, stable configuration and have made no attempt to understand if a numerically-induced instability at $a/W = 0.5$ remains unstable for other stable configurations $a/W > 0.5$. In contrast to the linear decay of [4], the potential of [21] contains a minimum in the unloading curve. We only comment that provided the cohesive zone is well resolved, one will always sample sufficient portions of the traction-displacement law to form an ill-conditioned stiffness matrix.

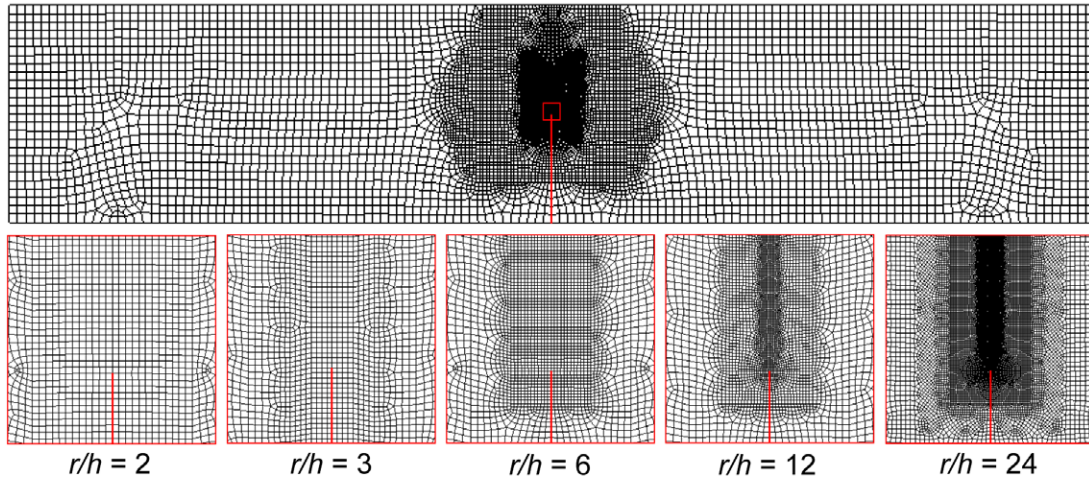


Fig. 4. Representative meshes employed for stability analysis. The cohesive surface element size h is normalized by the fracture length scale $r = E\phi/\sigma_{\max}^2$.

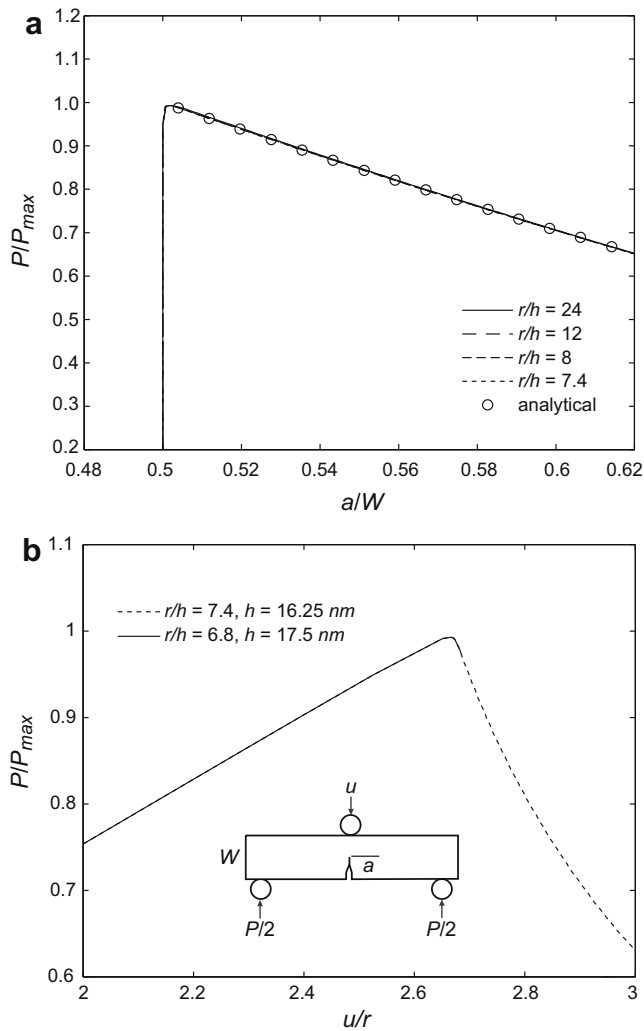


Fig. 5. Reduction in the normalized load $P/P_{\max}$ for (a) increasing normalized crack length a/W and normalized displacement u/r where $r = E\phi/\sigma_{\max}^2$. For all stable simulations (a), the cohesive zone is well resolved and all cases are converged. Decreasing the mesh size by ~ 1 nm, the system rapidly transitions from unstable to stable propagation (b).

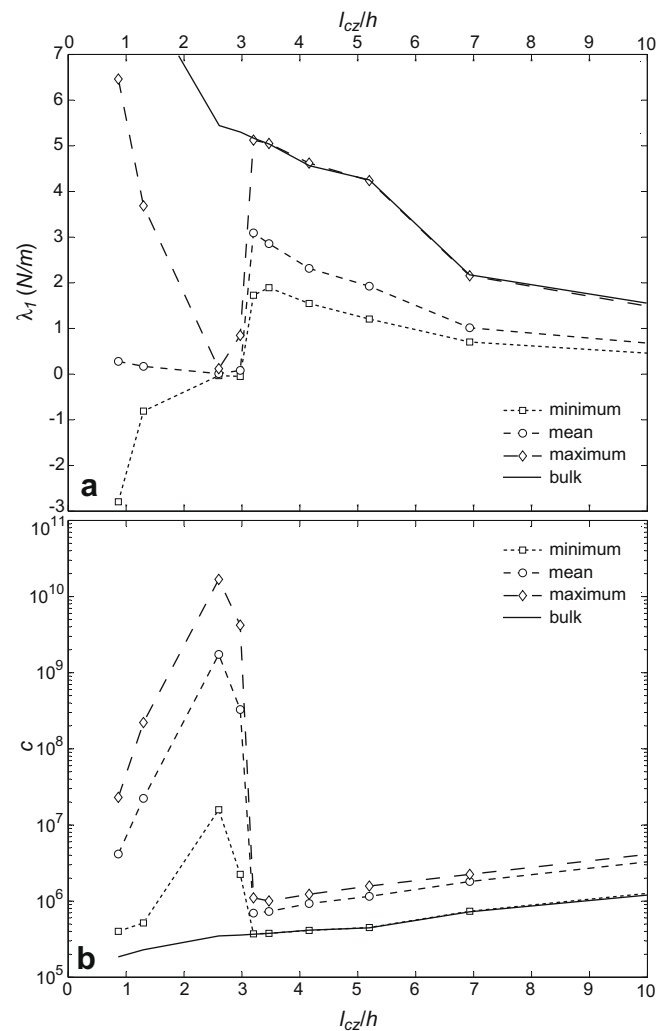


Fig. 6. The minimum, mean, and maximum (a) first eigenvalue λ_1 of the system and (b) condition number during the iterative process. The minimum eigenvalue illustrates the transition from indefinite to positive definite and the mean eigenvalue reflects a nearly singular system. Smaller mesh sizes $l_{cz}/h > 3$ significantly impact the condition number and ensure stability.

Another phenomena we did not explore rigorously is the role of the cohesive strength σ_{max} . Because the restriction in 1-D $h < \frac{1}{2} E\phi / \sigma_{max}^2$ provides the appropriate order of the mesh size, the trend in σ_{max} is clear. Decrements in the strength will aid the stability. We note that we could have easily expressed stability in terms of the characteristic length scale δ_n but the given relation is more familiar to the reader. We re-emphasize that if the bulk is elastic and the cohesive zone is contained, the only material property governing the resistance is the toughness ϕ [15]. For a single path, one does have some freedom to modify the cohesive strength to ensure stability. However, if multiple cracks (paths) are interacting, the cohesive strength will have a profound role [14,9] and one should employ refinement to determine if the instabilities in the simulation stem from crack interaction and not the mesh size.

5. Conclusion

Idealized cases in 1-D and 2-D provide illustrative cases to better understand quasi-static instabilities in the cohesive approach to fracture. Analytic expressions in 1-D provide a criterion for stability in terms of a fracture length scale $r = E\phi / \sigma_{max}^2$. Materials of small r , brittle materials, require increased discretization to ensure stability. Three-point bend simulations extend the concepts in 1-D and confirm that the element size required for crack stability can be less than the element size required for cohesive zone resolution.

If the discretization is insufficient, one can numerically induce an instability. We term the instability a numerical instability because it stems only from the selected mesh size. We make this claim because the presented three-point bend geometry is physically stable, $\partial J / \partial a < 0$. The delineation is needed because, in practice, we cannot evaluate $\partial J / \partial a$ analytically. If the cohesive zone l_{cz} is adequately resolved, one may associate a lack of convergence with a physically unstable system. An examination of the load–line force and displacement for $r/h = 6.8$ in Fig. 5b might confirm the presumption. For this case, an instability occurs just after unloading. However, if we decrease the mesh size by roughly a nm, $r/h = 7.4$, we obtain stable propagation.

Just as one refines the mesh to ensure that the cohesive zone is resolved, we propose that one may also need to refine the mesh to ensure that the solution is stabilized. Moreover, findings indicate that transition from indefinite to positive definite is abrupt and that small decrements in the mesh size greatly decrease the condition number and result in a global minimum for stable propagation. Further decrements in the mesh size align the condition with the bulk case and result in an increased condition number.

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